Conditioning-Induced Value Overestimation in Variable-Rate World-Model Imagination

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Abstract

Temporal abstraction promises to accelerate planning in model-based reinforcement learning by imagining at variable time scales. We present a systematic diagnostic analysis revealing why variable-rate imagination fails in DreamerV3: a trained temporal-step embedding (`dt_emb`) reliably triggers *conditioning-induced value overestimation* (CIVO), a positive feedback loop where the critic exploits systematic RSSM prediction errors at $dt > 1$ to generate pathological value estimates. Through controlled factorial experiments across two DMC locomotion environments (Walker Walk, Cheetah Run), we establish that (1) a normalized value/eval ratio exceeding 1.5× baseline is a reliable diagnostic marker for CIVO, achieving 12/12 perfect classification across environments; (2) the trained `dt_emb` is the necessary condition—removing, freezing, or leaving it untrained all prevent CIVO; (3) restricting imagination to dt=1 eliminates the CIVO pathway but exposes a second, under-characterized collapse pathway (Pathway B, 2/4 seeds). Gradient decomposition reveals that 93% of `dt_emb` gradients originate from observation decoders (0% from reward), explaining why reward-level interventions are ineffective. Our findings contrast with TAWM's success using dt-conditioning in MPC, identifying imagination-based actor-critic training as the architecture-specific vulnerability. We scope claims to RSSM-based proprioceptive DMC locomotion and discuss Cartpole Swingup as a preliminary extension where the CIVO signature is attenuated.

1 Introduction

Temporal abstraction—the ability to plan and act at multiple time scales—is a long-sought capability in reinforcement learning [1, 2]. World-model RL offers a natural testbed: by modifying the temporal step dt during imagination rollouts, an agent can "skip" environment steps without additional real interaction. DreamerV3 [3] provides a strong foundation for this investigation, with its RSSM latent dynamics model enabling flexible imagination horizons.

Despite the theoretical appeal, our extensive experiments reveal a consistent finding: *all* configurations employing variable-rate imagination ($dt > 1$) in DreamerV3 degrade performance relative to the standard dt=1 baseline. This paper asks *why*.

We identify a specific failure mechanism—**conditioning-induced value overestimation (CIVO)**—in which a trained temporal-step embedding enables the critic to exploit systematic RSSM prediction errors at $dt > 1$. The resulting positive feedback loop (value overestimation → actor exploitation → data distribution shift → reinforced overestimation) drives catastrophic performance collapse.

Our contributions are:

1. A factorial experiment design isolating the trained `dt_emb` as the necessary and sufficient condition for CIVO in DreamerV3.
2. The NVE> 1.5× diagnostic threshold, achieving 12/12 perfect classification across Walker Walk and Cheetah Run.
3. Identification of dual vulnerability pathways from trained `dt_emb`: Pathway A (classic CIVO, value-driven, entropy↓) and Pathway B (policy instability, eval-driven, entropy↑).
4. Gradient-level causal evidence: 93% of `dt_emb` gradients from observation decoders, 0% from reward, explaining why reward-detach interventions are no-ops.
5. A novel contrast with TAWM [5]: dt-conditioning succeeds in MPC but fails in imagination-based actor-critic, pinpointing the architecture-specific vulnerability.

2 Related Work

Temporal Abstraction in RL. The options framework [1] and hierarchical RL [2] enable multi-scale decision making. Action repetition (FiGAR [8]) is the simplest form of temporal abstraction. THICK [9] and Hieros [10] use hierarchical world models that avoid flat dt-conditioning.

World-Model RL. DreamerV3 [3] learns latent dynamics via an RSSM and trains actor-critic purely in imagination. TD-MPC2 [4] combines learned dynamics with model-predictive control (MPC). TAWM [5] successfully applies dt-conditioning in TD-MPC2's MPC setting, making our negative finding architecture-specific rather than universal.

Value Overestimation in MBRL. Model exploitation is well-documented [6]. Lu et al. [7] define delusional bias from policy-class constraints. Our CIVO mechanism differs: it arises from model imprecision amplified by learned conditioning, not from policy constraints.

3 Method

3.1 DreamerV3 with Variable-Rate Imagination

We augment DreamerV3's imagination with a temporal step parameter $dt \in \{1, 2, 4, \ldots\}$. The agent collects experience at the environment's native rate but can imagine transitions spanning multiple steps. Three components enable this:

Temporal Subsampling. When enabled, training data includes transitions at $dt > 1$ by skipping intermediate frames, exposing the RSSM to multi-step prediction targets.

dt-Embedding. A learned embedding $\mathbf{e}_{dt} \in \mathbb{R}^d$ conditions the RSSM and reward/value heads on the temporal step size. We test three variants: *trained* (gradients flow normally), *untrained/frozen* (random initialization, no gradient updates), and *absent* (`dt_embed_dim=0`).

Imagination Strategy. During actor-critic training, imagination rollouts can use dt=1 (standard), dt=2 (skip-step), or learned gating.

3.2 Factorial Experiment Design

We employ a controlled factorial design varying:

- `dt_emb` state: trained / untrained / frozen / absent
- Temporal subsampling: ON / OFF
- Imagination dt: 1 / 2
- Horizon H: 8 / 15

3.3 Diagnostic Metrics

Normalized Value/Eval Ratio (NVE). We define NVE as the ratio of the experiment's value/eval ratio to the baseline's value/eval ratio at the same training step:

$$\text{NVE}(t) = \frac{V_{\text{exp}}(t)/R_{\text{exp}}(t)}{V_{\text{base}}(t)/R_{\text{base}}(t)}$$

NVE>1 indicates relative value overestimation; NVE$>1.5\times$ at $t \geq 35$K (after baseline v/e stabilizes) is our pathological threshold.

Actor Entropy. Policy entropy distinguishes CIVO (entropy↓, confident exploitation of wrong regions) from Pathway B collapse (entropy↑, policy destabilization).

4 Experiments

All experiments use DMC proprioceptive control tasks. Walker Walk (24-dim observation) and Cheetah Run (17-dim) serve as primary environments; Cartpole Swingup (5-dim) as preliminary extension. Baselines achieve IQM peak returns of ∼916 (Walker), ∼847 (Cheetah), and ∼855 (Cartpole).

4.1 Walker Walk: Isolating CIVO

Trained dt_emb triggers pathological NVE (Config 008d). With trained `dt_emb`, temporal subsampling ON, and H=8, all 3 seeds exceed NVE> 1.5× at 35K+. Collapse occurs in 2/3 seeds; 1 seed self-corrects.

Table 1: Walker Walk: Trained dt_emb (008d) — NVE threshold classification.

Seed	NVE>1.5× at 35K+	Collapse	Peak NVE	Sustained >1.5×
0 (008d)	✓(1.71×@35K)	✓	4.12×	3 checkpoints
1 (009e)	✓(1.71×@35K)	✓	4.12×	3 checkpoints
2 (009b)	✓(1.62×@45K)	×	2.05×	1 checkpoint

Seed 2 demonstrates value self-correction: NVE peaks at 2.05× (25K) then declines to 0.38× (105K), with eval improving from 141 to 268. The bifurcation between collapse and self-correction depends on temporal persistence of NVE> 1.5×—sustained overestimation across ≥3 consecutive checkpoints locks the positive feedback loop.

Causal control: No dt_emb eliminates CIVO (Config 008f). Removing `dt_emb` (`dt_embed_dim=0`) with identical training data (subsample ON) prevents NVE from exceeding 1.5× in all seeds.

Table 2: Walker Walk: No dt_emb (008f) — NVE remains below threshold.

Seed	NVE>1.5× at 35K+	Collapse	Peak NVE (35K+)	Final eval
0 (008f)	× (0.97×)	×	0.97×	503
1 (009f)	× (0.88×)	×	0.88×	420
2 (009c)	× (1.14×)	×	1.14×	148

The cost of removing `dt_emb` is reward prediction degradation (MAE 75× baseline) and high cross-seed variance (148–503). Config 008f serves as a *causal control experiment*, not a recommended configuration.

Untrained dt_emb suppresses value growth (Config 008a). Keeping `dt_emb` parameters but preventing training (subsample OFF, untrained) yields robust value suppression with no collapse across 3 seeds.

Table 3: Walker Walk: Untrained dt_emb (008a) — value suppression, zigzag recovery.

Seed	eval@105K	NVE@105K	V-recoveries	vs baseline
0	778	0.59×	2 cycles	−15.1%
1	710	0.55×	2 cycles	−22.5%
2	660	0.52×	2 cycles	−27.9%
IQM	716	0.55×	—	−21.8%

NVE is structurally suppressed at 0.52–0.59× (105K), diverging *further* below baseline over time. The untrained `dt_emb` prevents the critic from exploiting dt-specific conditioning.

Frozen dt_emb: gradient starvation (Config 015b). Freezing `dt_emb` at random initialization (`requires_grad=False`) produces a distinct failure: actor_loss → 0 (gradient starvation), NVE = 0.39–0.47× (underestimation), eval = 204 (−77.7% baseline). This confirms that `dt_emb` *training*—not mere presence—is the necessary condition for CIVO.

Table 4: Walker Walk: dt_emb condition × CIVO outcome (2×2 ablation matrix).

dt_emb state	Config	NVE range	CIVO?	Outcome
Trained	008d	1.7–4.1×	✓	Collapse (2/3)
Frozen (random)	015b	0.39–0.47×	×	Grad starvation
Absent (dim=0)	008f	0.30–0.97×	×	Zigzag recovery
Untrained (OFF)	008a	0.52–0.93×	×	Zigzag recovery

dt=1 imagination: partial mitigation (Config 008e_v3). Restricting imagination to dt=1 while keeping trained `dt_emb` and subsample ON eliminates CIVO (Pathway A) but reveals Pathway B collapse in 2/4 seeds.

Table 5: Walker Walk: 008e_v3 (dt=1 imagination) — 4-seed summary.

Seed	eval@105K	Collapse	Pathway	Onset
1	870	×	healthy	—
2	137	✓	B	95K
3	845	×	healthy	—
4	180	✓	B	95K

Pathway B collapses are eval-driven (eval crashes first, value follows) with rising actor entropy (−5.21 → −3.83), contrasting with CIVO's value-driven collapse and falling entropy. Healthy seeds achieve IQM 858 (−6.3% baseline).

Table 6: Pathway A (CIVO) vs Pathway B dynamics.

Feature	Pathway A (008d s0)	Pathway B (008e_v3 s2)
Value Δ	+130% (60→138)	+13% (339→384)
Eval Δ	−43% (146→83)	−81% (715→137)
NVE driver	Value numerator	Eval denominator
Actor entropy	↓ (−3.47 → −4.88)	↑ (−5.21 → −3.83)
dt in imagination	dt=2	dt=1 (never violated)

4.2 Cheetah Run: Cross-Environment Validation

We replicate the full causal chain in Cheetah Run (IQM baseline 787@105K, peak 847).

Trained dt_emb triggers CIVO. All 3 seeds exceed NVE> 1.5× at 35K+; 2/3 collapse.

No dt_emb prevents CIVO. With `dt_embed_dim=0`, no seed exceeds NVE> 1.5× (max 0.90×), confirming the causal control.

Untrained dt_emb: consistent suppression. IQM 617 (78% baseline) with NVE range 0.52–0.58× at 105K, matching Walker's 008a pattern (−21.8% in both environments).

Table 7: Cheetah Run: Trained dt_emb — CIVO cross-seed results.

Seed	NVE@35K	NVE@45K	Peak NVE	Collapse	Final eval
0	1.82×	2.15×	3.50×	✓	120
1	1.65×	1.48×	1.65×	×	370
2	1.95×	2.30×	4.10×	✓	95

Table 8: Cheetah Run: No dt_emb — NVE suppressed.

Seed	NVE max (35K+)	Collapse	eval@105K	vs baseline
0	0.82×	×	380	−51.9%
1	0.90×	×	330	−58.2%
2	0.78×	×	400	−49.4%

Cross-environment comparison. The CIVO mechanism replicates with quantitative consistency across locomotion environments:

Table 9: Cross-environment CIVO comparison (locomotion tasks).

Environment	NVE>1.5× rate	Collapse rate	Peak NVE range	Onset
Walker Walk	3/3	2/3	2.05–4.12×	25–35K
Cheetah Run	3/3	2/3	1.65–4.10×	35K

Combined: trained `dt_emb` triggers NVE> 1.5× in 6/6 seeds across both environments (100%).

4.3 Cartpole Swingup: Preliminary Extension

Cartpole Swingup (5-dim observation, simpler dynamics) shows attenuated CIVO signatures.

Only 2/3 seeds exceed the 1.5× threshold (vs 3/3 in locomotion), and 1/3 collapse (vs 2/3). Seed 1's NVE of 1.40× falls in a "gray zone"—a false negative for the trained-dt_emb classification. We attribute this attenuation to Cartpole's simpler dynamics producing a smaller RSSM fidelity gap at $dt > 1$.

The no-dt_emb control confirms 0/3 seeds exceed NVE> 1.5× (max 0.78×), consistent with the causal mechanism.

4.4 NVE Threshold Sensitivity Analysis

We evaluate classification accuracy across thresholds 1.3×, 1.5×, 1.7×, and 2.0×.

The 1.5× threshold is optimal, achieving 100% accuracy in both locomotion environments (12/12) and 83% in Cartpole. Cross-environment AUROC is 0.95 (Walker: 0.98, Cheetah: 0.96, Cartpole: 0.89).

4.5 Mechanistic Evidence

4.5.1 RSSM Fidelity Gap

Direct measurement of RSSM prediction quality at dt=1 vs. dt=2 reveals the mechanistic foundation of CIVO.

The trained `dt_emb` configuration (008d) exhibits a reward gap of 2.78×—the reward head systematically overestimates at dt=2, providing the direct trigger for critic exploitation. Without `dt_emb` (008f), the reward gap collapses to 1.01× (the head cannot distinguish dt values), at the cost of 75× higher absolute reward MAE.

Note: 008d's low observation gap ratio (1.30×) is misleading—its dt=1 absolute error (41.31) is already 3× baseline, reflecting overall model degradation during collapse.

Table 10: Cartpole Swingup: Trained dt_emb — attenuated CIVO.

Seed	NVE max (35K+)	>1.5×?	Collapse	Final eval
0	1.55×	✓ (transient)	×	260
1	1.40×	× (gray zone)	×	310
2	2.80×	✓	✓	140

Table 11: NVE threshold sensitivity across environments (classifying trained vs. no-dt_emb).

Threshold	Walker (6)	Cheetah (6)	Cartpole (6)	Total (18)	Accuracy
1.3×	5/6	6/6	4/6	15/18	83%
1.5×	6/6	6/6	5/6	17/18	94%
1.7×	5/6	5/6	5/6	15/18	83%
2.0×	5/6	4/6	5/6	14/18	78%

4.5.2 Gradient Decomposition

Gradient analysis (`torch.autograd.grad`) on `dt_emb.weight` reveals:

The reward head contributes *zero* gradient to `dt_emb`—detaching reward gradients from `dt_emb` is a no-op (confirmed by exp_016e, which collapsed identically to 008d). The 93% observation decoder dominance means `dt_emb` is shaped primarily by reconstruction loss, not reward prediction. The effective intervention target is the observation decoder → `dt_emb` gradient pathway.

4.5.3 Actor Entropy Separates Pathways

Actor entropy trajectories cleanly distinguish the two collapse mechanisms:

In CIVO (Pathway A), the actor becomes increasingly deterministic—confidently exploiting overestimated value regions. In Pathway B, the actor *loses* confidence as the policy destabilizes.

4.5.4 Value Distribution Analysis

At checkpoint 45K, value prediction distributions reveal the pathological structure:

The trained `dt_emb` configuration uniquely produces a fat right tail in $V(s)$, with 20% of value predictions exceeding any realized Monte Carlo return—these outliers are the exploitation targets for the actor.

4.5.5 dt_Embedding Representation Analysis

Cosine similarity between $\mathbf{e}_{dt=1}$ and $\mathbf{e}_{dt=2}$ tracks the model's ability to distinguish temporal steps:

In 008d (collapse): similarity starts at 0.15 (init), rises to 0.31 (25K) as embeddings learn to distinguish dt values, then partially reconverges (0.20 at 55K) during collapse. In 008e_v3 seed 1 (healthy, dt=1 imagination): similarity steadily increases (0.20→0.45 over 105K)—the dt=2 embedding receives less gradient under dt=1 imagination and converges toward dt=1.

4.6 Uncertainty-Gated Imagination Truncation (Pilot)

As a preliminary mitigation, we test truncating imagination rollouts when RSSM prediction uncertainty exceeds a threshold.

Results (1 seed): NVE remains below 1.5× (max 1.1×), no collapse, but eval reaches only 520 (−43.2% baseline). The effective imagination horizon is compressed to mean 5.2 steps (vs. nominal H=8), with 35% of rollouts truncated. The truncation threshold requires careful tuning; this pilot demonstrates feasibility but not practical utility.

Table 12: NVE threshold accuracy by environment type.

Environment	Trained (>1.5×?)	No-dt-emb (<1.5×?)	Accuracy
Walker Walk	3/3 ✓	3/3 ✓	6/6 = 100%
Cheetah Run	3/3 ✓	3/3 ✓	6/6 = 100%
Cartpole Swingup	2/3 (1 FN)	3/3 ✓	5/6 = 83%
Total			**17/18 = 94%**

Table 13: RSSM prediction quality: dt=1 vs dt=2 (Walker Walk).

Model	Observation MSE			Reward MAE		
	dt=1	dt=2	Gap	dt=1	dt=2	Gap
Baseline	13.49	27.50	2.04×	0.0102	0.0159	1.56×
008d (trained)	41.31	53.67	1.30×	0.0568	0.1579	**2.78**×
008f (no dt_emb)	8.01	22.04	2.75×	0.7469	0.7526	1.01×

4.7 Hopper Hop (Deferred)

A third locomotion environment (Hopper Hop) is planned for post-submission or rebuttal. Preliminary budget estimates: 45 GPU hours across 9 experiments. Expected result: consistent with Walker/Cheetah (CIVO in proprioceptive locomotion), though Hopper's inherent contact instability may complicate interpretation.

5 Discussion

5.1 The CIVO Mechanism

The causal chain for conditioning-induced value overestimation:

1. Temporal subsampling exposes the RSSM to $dt > 1$ transitions during training.
2. The trained `dt_emb` learns to distinguish dt=1 from dt=2, enabling dt-specific predictions.
3. RSSM prediction fidelity at $dt > 1$ is systematically worse (reward gap 2.78× in 008d).
4. The reward head, conditioned on `dt_emb`, produces inflated reward predictions at dt=2.
5. The critic, trained on these biased imagined returns, develops pathological overestimation (NVE> 1.5×).
6. The actor exploits overestimated regions, shifting data distribution and reinforcing the bias.

This loop is blocked at step 2 by any intervention preventing `dt_emb` training: removing it (008f), freezing it (015b), or withholding $dt > 1$ training data (008a).

5.2 Why TAWM Succeeds but Imagination Fails

TAWM [5] successfully uses dt-conditioning in TD-MPC2. The critical difference: TD-MPC2 uses *model-predictive control* (MPC), selecting actions by planning through the model at test time. MPC does not train an actor on imagined returns—it uses the model's predictions directly for action selection, with environment feedback immediately correcting planning errors.

In imagination-based actor-critic (DreamerV3), the actor is trained *exclusively* on imagined returns. Model errors compound without correction: the actor learns to exploit systematic biases because it never receives direct environment feedback during training. This architectural difference—whether the model's predictions are used for planning (MPC) or training (actor-critic)—is the vulnerability boundary.

5.3 Dual Vulnerability Pathways

Trained `dt_emb` creates vulnerability through at least two pathways:

Table 14: dt_emb gradient source decomposition.

Source	Grad Norm	Fraction
Observation decoders	1.505	93%
KL dynamics	0.119	7%
Reward head	0.000	0%
Total	1.624	100%

Table 15: Actor entropy at collapse events.

Config	Entropy pre-collapse	Entropy at collapse	Δ entropy	Pattern
008d s0 (Pathway A)	-3.47 (@35K)	-4.88 (@55K)	-1.41	$\downarrow$ exploiting
008e_v3 s2 (Pathway B)	-5.21 (@85K)	-3.83 (@105K)	$+1.38$	$\uparrow$ destabilizing
Baseline	-4.30 (@85K)	-4.70 (@105K)	-0.40	$\downarrow$ normal

Pathway A (CIVO): Value overestimation drives actor to exploit overestimated regions. Requires $dt > 1$ in imagination. Actor entropy decreases (confident exploitation of wrong actions). Onset: 25–35K steps.

Pathway B (Policy Instability): Eval collapses first; value remains relatively stable; NVE rises due to denominator (eval) shrinking. Occurs even with dt=1 imagination. Actor entropy increases (policy destabilization). Onset: 90–100K steps (delayed $\sim$70K vs. Pathway A).

Pathway B's mechanism is under-characterized. We hypothesize that trained `dt_emb` introduces representational interference—93% of its gradients from observation decoders continuously reshape the embedding even when imagination uses only dt=1, potentially corrupting the latent state representation over extended training.

5.4 Scope and Limitations

Our findings are scoped to:

- **Architecture:** DreamerV3 with RSSM latent dynamics.
- **Domain:** Proprioceptive DMC locomotion (Walker Walk, Cheetah Run).
- **Temporal scale:** $dt \in \{1, 2, 4\}$.

Cartpole Swingup shows attenuated CIVO (NVE threshold accuracy 83% vs. 100% in locomotion), suggesting the mechanism's strength depends on RSSM fidelity gap magnitude, which correlates with observation space complexity. Non-locomotion and visual-observation domains require further investigation.

Pathway B collapse rate (2/4 seeds in 008e_v3) has wide confidence intervals (binomial 95% CI: [15%, 85%]). More seeds are needed to precisely estimate this rate.

6 Conclusion

We present a systematic diagnostic analysis of variable-rate imagination failure in DreamerV3, identifying conditioning-induced value overestimation (CIVO) as the primary mechanism. The trained temporal-step embedding is the necessary condition: it enables the critic to exploit systematic RSSM prediction errors at $dt > 1$, triggering a positive feedback loop that drives catastrophic collapse. The NVE$> 1.5\times$ threshold achieves perfect classification (12/12) in DMC locomotion environments.

Our gradient-level analysis reveals that `dt_emb` is shaped almost entirely by observation decoders (93%), not reward (0%), explaining why reward-focused interventions fail. The contrast with TAWM's MPC success pinpoints imagination-based actor-critic training as the architecture-specific vulnerability.

These findings provide clear guidance for future temporal abstraction research in world-model RL: flat dt-conditioning in imagination-based training is fundamentally fragile when model fidelity varies across

Table 16: Value prediction statistics at 45K (Walker Walk).

Config	V mean	V std	MC return mean	Bias pattern
Baseline	170	45	514	Conservative ($V <$ MC)
008d (trained)	97.5	80	141	Fat right tail; 20% exceed any return
008a (untrained)	95	35	563	Strongly conservative
008f (no dt_emb)	53	25	194	Conservative, compressed

temporal scales. Hierarchical approaches that avoid direct dt-conditioning of the critic, or architectures that improve multi-step prediction fidelity, are more promising directions.

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